

Trained

Place labels decide what counts



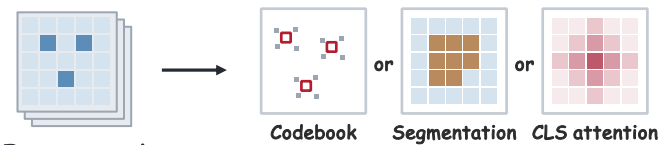
Large labelled
training set

Corpus-specialized
features

ID Strong within-domain discrimination
OOD Can inherit training-domain bias

Training-free, prior work

Added criteria decide what counts



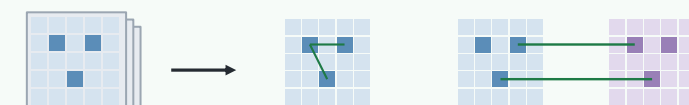
Frozen patch
features

Prior-selected / aggregated features

ID Selection ≠ place discrimination
OOD Selection criteria may not transfer

Ours

Let patches decide what counts



Frozen patch
features

Within-image

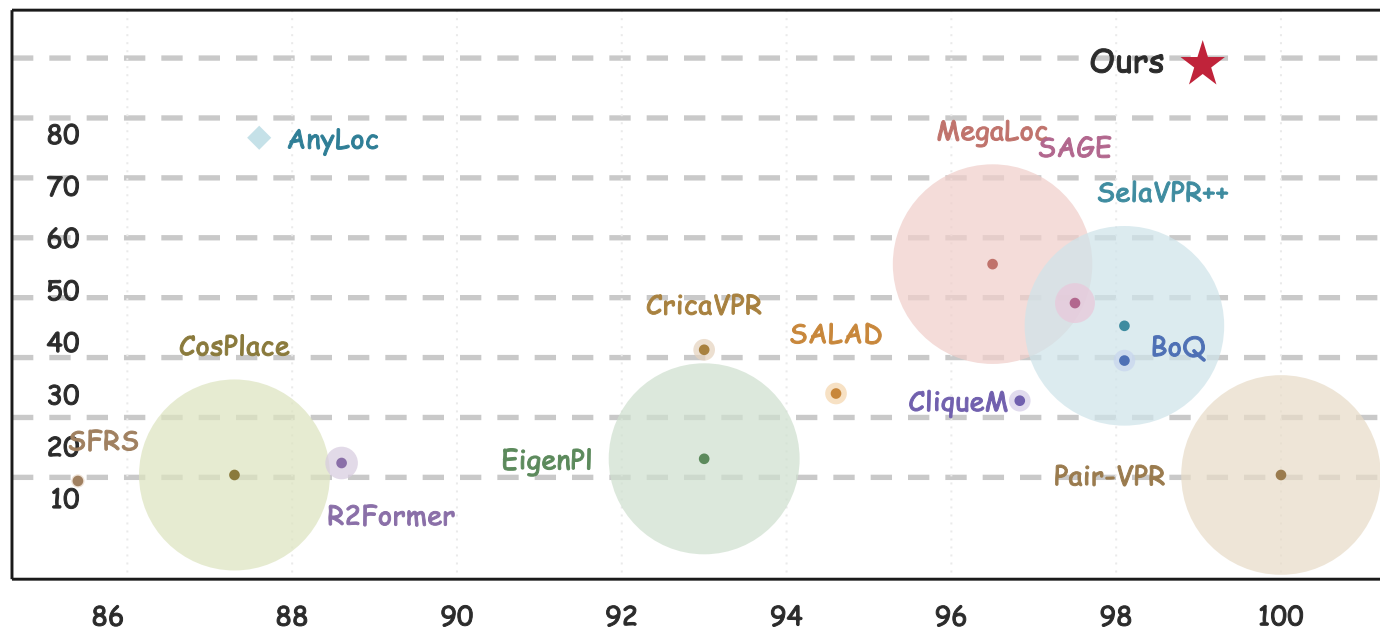
Across-image

Balanced, matchable patch evidence

ID Match distinctive local evidence
OOD Strong cross-environment retrieval

Supervision cost vs. OOD recall

VP-Air R@1 (out-of-distribution, %)



★ ours (training-free) ♦ training-free ● trained ● 1 million images ● 5 million images

Recall across 14 benchmarks

